(Please write your Enrollment Number) Enrollment No. 06011702024

# End-TermExamination
## (CBCS)(SUBJECTIVE TYPE)(OffLine)
## Course Name:< B.Tech >, Semester:< 6th Sem >
## (May, 2024)

| Subject Code: BCS 306 | Subject: Compiler Design |
|---|---|
| Time :3 Hours | Maximum Marks :60 |

**Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.**

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|---|---|---|
| **Q1** | | **(2.5*8=20)** |
| | (a) Write about the first phase in Compiler? How a token is recognized, explain with an example? | |
| | (b) What is the importance of semantic analysis, explain? | |
| | (c) What is top down parsing? Design Recursive Descent Parser for the following grammar:<br>S→AA, A→aA, A→b | |
| | (d) Briefly explain about Syntax Directed Translation. | |
| | (e) What is code optimization and how it is useful in compilation process? | |
| | (f) Define FIRST and FOLLOW function Rules, explain with an example Grammar. | |
| | (g) What is the second phase in Compiler? How grammar is useful in second phase? | |
| | (h) Define Ambiguity? Check whether the grammar S→aAB, A→bC/cd, C→cd, B→c/d, is Ambiguous or not? | |
| | **UNIT-I** | |
| **Q2** | With a neat diagram discuss about different phases of the compiler? Explain with suitable example. | **(10)** |
| **Q3** | Briefly explain about the following with suitable example:<br>(i) Finite Automata<br>(ii) Derivation tree and parse tree<br>(iii) Ambiguity with an example<br>(iv) Bootstrapping | **(10)** |
| | **UNIT-II** | |
| **Q4** | Discuss about SLR parser? Construct SLR parse table for following grammar.<br>E→E+T<br>E→T<br>T→T*F<br>T→F<br>F→(E)<br>F→id | **(10)** |
| **Q5** | By explaining the steps followed to construct LALR parse table, Design a LALR parse table for the following grammar.<br>S→AA<br>A→aA/b | **(10)** |
| | **UNIT-III** | |
| **Q6** | Define what Syntax directed definitions is, and explain about following with example.<br>(i). Bottom-up evaluation of S- attributed Definitions<br>(ii). Bottom-up evaluation of inherited attributes. | **(10)** |
| **Q7** | (i). Discuss about different Source Language issues?<br>(ii). Explain about various Storage allocation strategies? | **(10)** |
| | **UNIT-IV** | |
| **Q8** | Explain about the following with suitable example:<br>(i). Three Address code<br>(ii). Quadruples<br>(iii). Triples<br>(iv). Graphical Representation of Intermediate language. | **(10)** |
| **Q9** | What is code optimization, What are its uses? Explain in detail about Principal sources of optimization with suitable example. | **(10)** |